

Tunisia, audited — the state program, complete

The Fed transmission measured, the program architecture, the instruments, the financing, the gates, the signature path — every line cited.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

1. The Fed transmission — measured, not restated

World Bank API, measured 2026-09-09 (Pg1): external debt **\$40.46B**, **78.7% of GDP**; debt service **26.2% of export revenues**; reserves **\$9.34B** = 4.3 months of imports; food \$3.00B + fuel \$4.90B of imports; CA -1.5% of GDP; growth 2.5%, inflation 5.2% (2025).

Channel 1 — the debt trap: the “original sin” (Pg12) — no long-term borrowing in the dinar abroad — exposes the state to the Fed cycle (Pg13): Fed hikes lift global yields, refinancing turns expensive, and the 26.2% of export revenues that debt service absorbs compounds while reserves (4.3 months) drain.

Channel 2 — imported inflation: the rate differential pulls capital to US safe assets, the dinar depreciates, and the dollar invoice on food (\$3.00B) and fuel (\$4.90B) inflates. The BCT — autonomous, non-monetizing under Organic Law 2016-35 (Pg4/Pg5) — must hold rates to defend the dinar: correct orthodoxy, strangled growth.

The loop’s answer: owned export engines earn ~\$117M/yr of gross exports outside the Fed cycle; the substitution legs take \$152M/yr out of the dollar invoice directly; reserves rise, the rating runs B-/Caa1 → BB-, and the refinancing premium shrinks for the state with its own hard-currency engines. Not defiance of the Fed — independence from the channel.

2. The state program — architecture, instruments, financing, adoption

Counterparts — who signs what

INSTITUTION	INSTRUMENT	ROLE
GCT (Tunisian Chemical Group)	acid-stream MOU + sampling agreement	the treasury's feedstock
SONEDE	municipal/industrial water offtake at >=\$0.68/m3 gate	water contracts under the existing regime
STEG	25-year PPA under the IPP regime (500 MW class)	the power leg's revenue
Ministry of Industry, Mines & Energy	program sponsor; tender registration	the executive mandate
Ministry of Health	biomonitoring + cohort registry	the scoreboard
CNSTN	NORM licensing for the uranium route	the regulatory pass
Ministry of State Domains & Agriculture	40-year leases under Law 93-120 (86,000 ha available)	the land
BCT	reserve/rating path; FX policy coordination	the monetary endgame

Legal instruments — nothing new needs to be invented

INSTRUMENT	WHAT IT PROVIDES	CITATION
Organic Law 2016-35 (BCT)	monetary autonomy; the program does NOT monetize	Pg4/Pg5
IPP regime — Law 2015-62 & 2024/25 awards	renewables concessions; ~\$31/MWh PPAs; 500 MW precedent	Rc13
PPP framework — Law 2015-49	desal/metals as PPPs where the state is the offtaker	Pg2
Law 93-120 (state-domain leases)	40-year agricultural leases; ~86,000 ha earmarked	Rc19
NORM/radiological regime (CNSTN, IAEA standards)	the uranium route's license layer	Pg11/Rm12

Financing stack — staged, gated, modeled in code

STAGE	FINANCING	GATE / CONDITION
Phase 0 (\$150-253k)	grant / founder / early mandate	no DFI needed; 90 days
Health pilot (\$58k)	ESG/grant	the first scoreboard numbers
Metals plant (~\$100M)	60% debt @7% + offtake security; MIGA	DSCR >=1.2 at contract prices (nations_v5)
Solar (500 MW, ~\$550M)	DFI stack: EIB/AfDB/WB-IFC at 4.5% + MIGA; or lean build	DSCR gate inside the bankable region (nations_v5)
Desal (~\$165M)	70% debt @6% + municipal offtake + sovereign backstop	blended >=\$0.68/m3, >=60% pre-sold
Terfas + food (\$7-35M)	concessional agri + founder	nursery first
EU Global Gateway (17 GW)	the national envelope this program plugs into	Pg7

Governance

- every gate is a public number — published, dated, re-checkable
- health fund separate from the industrial SPVs; one shared data platform
- quarterly scoreboard: water chemistry, filter uptime, cohort KPIs, price index in Gafsa/Gabes
- each pilot passes or dies alone, in public — the falsification ladder is the governance
- commercially-validated ledger published at zero until the first signature moves it

Adoption path — what the state signs, in order

1. **Month 0:** the state signs: GCT sampling MOU + Phase 0 mandate (\$150-250k, 90 days, no construction)
2. **Day 90:** two anchor measurements land: the 100-point water panel and the acid-stream assays — every projection becomes a signed number
3. **Months 4-18:** pilots on public gates: acid pilot, 100 ha terfas + nursery, spirulina pond, solar bid, desal offtake, health KPIs
4. **Years 2-5:** build only what passed: metals, solar, desal, health at basin scale — module by module, each with its own financing
5. **Years 5-10+:** the full loop: municipal spine, compute colocation, hydrogen, brine valorization — the destination the original design described

3. The household ledger — what this does to the cost of living

A state program that does not make life cheaper for its people is decoration. Each line, computed from the same primitives as the rest of this document; the quarterly price index in Gafsa and Gabès is the published enforcement: **the price of protein and vegetables must bend downward within 24 months of the first harvest.**

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3. The energy ledger — what power costs, per party

Anchors cross-checked: Rw10 (0.352 TND/kWh industrial, 19 TWh), Pg14 (the government’s own 1,100 GWh = 5% of production, ~\$125M/yr gas saving), Rc13 (\$31/MWh inside the \$24-35 historical bid band), Pg15 (fuel imports \$4.895B, WITS-verified).

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3. The frontier — where the loop reaches, and where it honestly stops

The complete household budget, swept — nothing hidden, nothing promised beyond the frontier.

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The water stack — line-by-line, with each credit's failure mode

The \$0.347/m³ audit: the base is measured anchors; each credit is a loop-internal yield with a named failure mode; the degraded cascade still clears the \$0.68 gate.

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LINE	\$/M³	PROVENANCE	WHAT MAKES IT FAIL
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LINE	\$/M³	PROVENANCE	WHAT MAKES IT FAIL
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The AI runs the measurement, the routing and the publication — never the decisions.

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The results framework — what a World Bank board reads first

The scoreboard IS the M&E system (PforR class, Pg23).

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The people's energy is electrons, not hydrogen — and the H&sb2; lives inside the loop

The loop's H&sb2; (2.63 kt/yr) is an industrial molecule: per kWh of heat, green H&sb2; at \$0.21 is 7× the loop's \$0.031/kWh electron; blending caps at 5-20% by volume (Pg20/Pg21/Pg22); the whole H&sb2; volume equals 0.0% of grid-gas demand. The molecule goes where only a molecule works: ammonia for the GCT chain, inside the loop. the electrolyzer sits on the desal line, and everything it touches stays in the loop: the water it splits is the SWRO permeate (0.04% of the 150,000 m³/d output); its un-split reject stream is clean water that returns to the product line; its oxygen by-product aerates the spirulina ponds and greenhouses; its stack heat (15–20% of input) joins the kiln and DC reject heat on the desal preheat circuit; and the minerals concentrated in its flush stream join the brine valorization line. The electrolyzer is not a consumer of the loop — it is another recovery point inside it.

4. What the state receives

External account: ~\$279M/yr of exports + import substitution + financing delta at full phase — measured against the 26.2% of export revenues that debt service takes today. **Rating path:** B-/Caa1 → BB- and reserves 4.5 → 6+ months as the owned-export engines compound (Rc11). **Health scoreboard:** the basin's water chemistry, DALY ledger and filter uptime become public KPIs — the collateral that lowers the price of every loan. **Jobs:** workshop, nursery, ponds, plants, solar crews, health workers — in the towns where the 2008 and 2015 blockades began. A job is the cheapest peace treaty ever signed.

5. The gates are public — the state sees every number

MODULE	GATE	PASSES ON
Health pilot (100 households, 2 schools, 90 days)	filter uptime ≥70%, urinary F ↓20%	public KPIs
Metals pilot	U recovery ≥85%, all-in ≤\$75/lb, offtake signed	GCT stream
Solar	DSCR ≥1.15 inside the bankable region, or grant support	PPA award
Desalination	≥60% contracted offtake, blended ≥\$0.68/m³	brine permit
Terfas	3-season record ≥200 kg/ha at ≥€20/kg; quota clause signed	scaling toward 2,000-3,000 ha

6. Phase 0 — what the 90 days produce for the state

Two anchor measurements — the 100-point water panel and the acid-stream assays (\$75k of the \$253k list) — plus the emissions inventory. Every projection on this page becomes a signed number before one brick is laid. Nothing is asked of the state beyond cooperation, and every gate is a number the state can check.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

Works cited

1. [Rm17] GCT throughput (~6.5 Mt phosphate rock/yr; >10 Mt/yr PG at peak; 'hundreds of millions of tons' discharged at Gabès) — http://www.gct.com.tn/fileadmin/user_upload/Downloads/Rapport_Annuels/GCT-ANNUAL-REPORT-2012-PART1.pdf...

2. [Rw7] Tunisian water tariffs: SONEDE 0.200-2.310 TND/m³; irrigation 0.060-0.160 TND/m³; groundwater 0.112-0.162 TND/m³ — <https://www.sonede.com.tn/tout-savoir-sur-leau/la-facture-deau-methodes-de-paiement/...>

3. [Rw10] Tunisia grid: STEG 5.9 GW, ~19 TWh/yr, 95% gas-fired, renewables <5%, industrial tariff 0.352 TND/kWh; budget deficit 5-7% GDP, debt ~83% GDP — <https://www.trade.gov/country-commercial-guides/tunisia-electrical-power-systems-and-renewable-energy...>

4. [Rw12] Tunisia land law: foreign ownership of agricultural land prohibited; 40-year leases only; state-domain conversion is an administrative hurdle — <https://2009-2017.state.gov/e/eb/ifa/2008/101021.htm...>
5. [Rc11] Tunisia sovereign ratings: Moody's Caa1 (Feb 2025 upgrade, stable); Fitch B- (Sep 2025, affirmed Jan 2026); reserves ~4.5 months import cover; WACC 10-15% — <https://www.fitchratings.com/research/sovereigns/fitch-upgrades-tunisia-to-b-outlook-stable-12-09-2025...>
6. [Rc12] Who finances Tunisia: EIB, AfDB, KfW, World Bank/IFC, EU Global Gateway (17 GW renewable programme); no active IMF program — <https://international-partnerships.ec.europa.eu/policies/global-gateway/17-gw-renewable-energy-programme-tunis...>
7. [Rc13] Empirical IPP record: 500 MW solar awarded 2024/25 (Scatec, Qair, Voltaia 25-yr PPAs at ~\$0.031/kWh); installed renewables ~400 MW; STEG debt > \$1.3B — <https://www.engineeringnews.co.za/article/scatec-awarded-ppas-for-wind-solar-projects-in-tunisia-2026-01-26...>
8. [Rc19] Tunisia arid land: ~86,000 ha of state domain earmarked for 40-year agricultural leases (Law 93-120) — <https://documents1.worldbank.org/curated/en/926441468778187741/pdf/multiopage.pdf...>
9. [Ru2] URANEXT/OCF (Morocco) building the first modern yellowcake-from-phosphoric-acid plant, ~\$100M CAPEX, El Jadida, 2025 — <https://northafricapost.com/87963-uranext-to-build-moroccos-first-yellowcake-uranium-plant-in-el-jadida.html...>